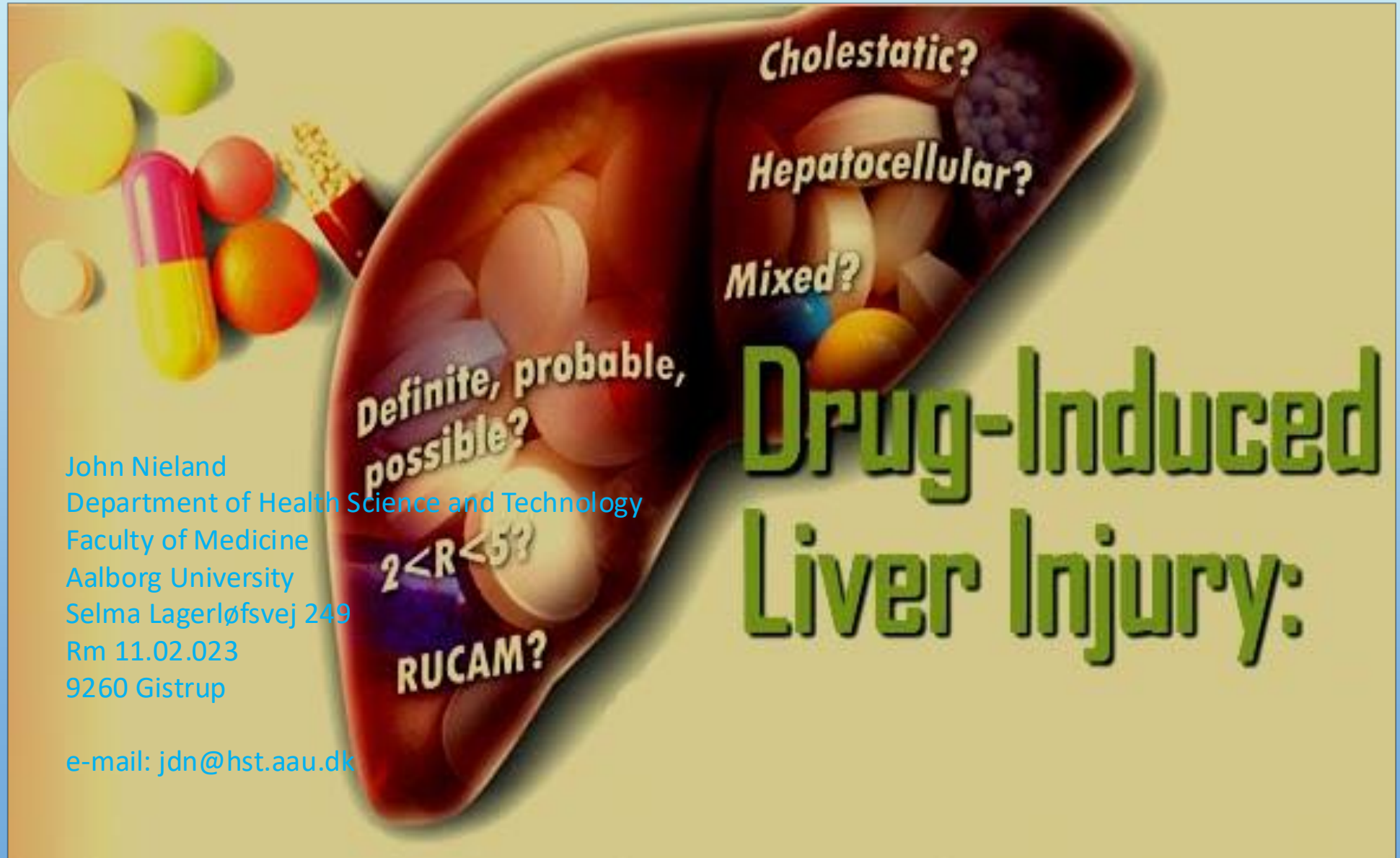
Drugs and the liver



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LEARNING OUTCOMES

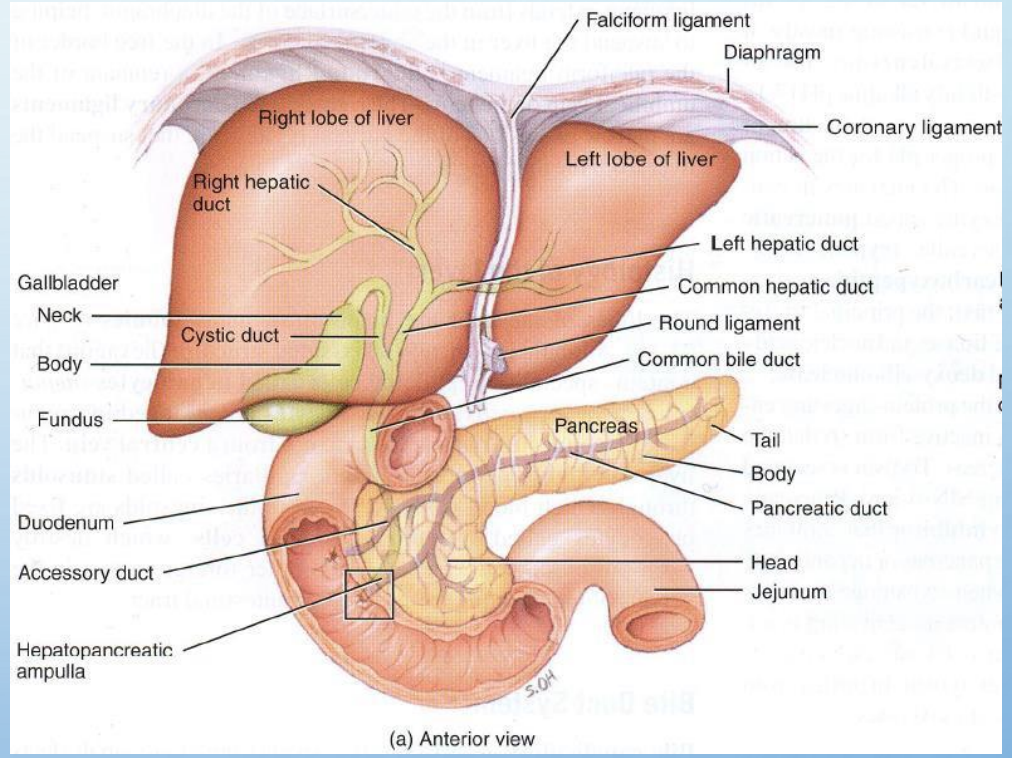
- Role of the liver in drug metabolism
- Role of drugs in liver damage
- Consequences of hepatic damage to drug metabolism & excretion

LIVER & BILIARY SYSTEM

Metabolic & secretory functions

- Plasma protein synthesis (e.g. albumin, steroid-binding proteins)
- Bile formation & secretion
- Metabolism of nutrients & vitamins
- - inactivation
  - activation
  - detoxification
  - excretion

  … of various substances



Jenkins/Kemnitz/Tortora, Anatomy & Physiology, Wiley 2010

DRUG EXCRETION

Only hydrophilic molecules excreted effectively by the follwing routes:

- Kidney
  quantitatively most important, involving processes:
  - glomerular filtration
  - tubular secretion
  - tubular reabsorption
- Bile
  - active secretion of drug metabolites from hepatocytes into bile
  - enterohepatic cycling: deconjugation in the GIT due to pH and enzymes
    → reabsorption into the plasma (long drug duration)
- Sweat
- Lungs
  - gaseous lipophilic molecule (inhalation anaesthetics)
- Breast milk
  - lower pH of breast milk than plasma:
    → basic drugs (ion/pH trapping)

DRUG ELIMINATION

Main processes

- **Metabolism/Biotransformation**
  enzymatic conversion of one chemical entity to another within the body
- **Excretion**
  elimination from the body of chemically unchanged drug or its metabolite

Main routes

- kidney (urine)
- hepatobiliary system
- lungs

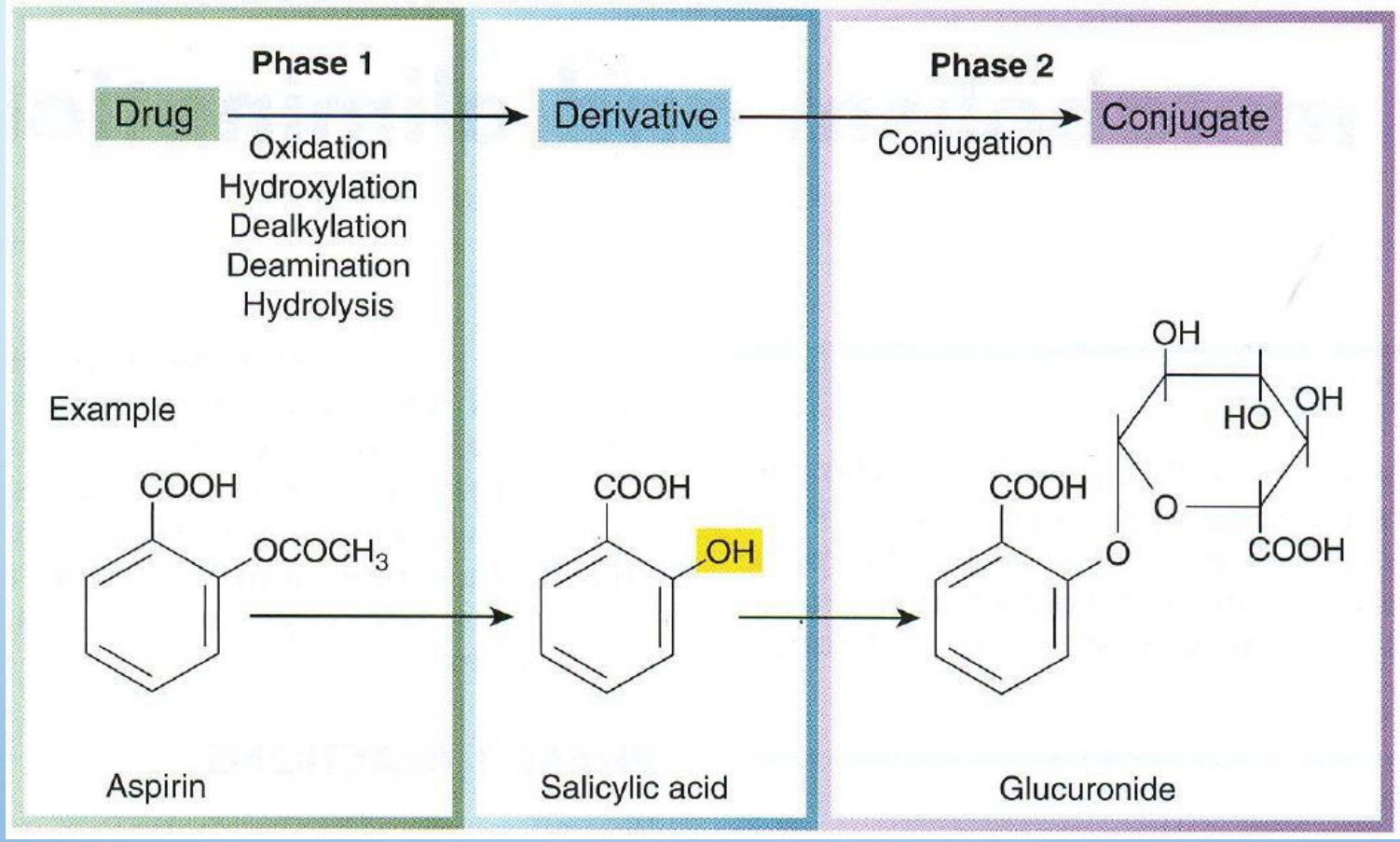
METABOLISM

- Drugs with small molecular volumes or polar characteristics
  → mainly: **renal excretion**

- Lipophilic drugs are eliminated via kidney inefficiently
  → lipophilic drugs will have prolonged duration & action
  → **metabolized** to more polar products
  → **urinary excretion**

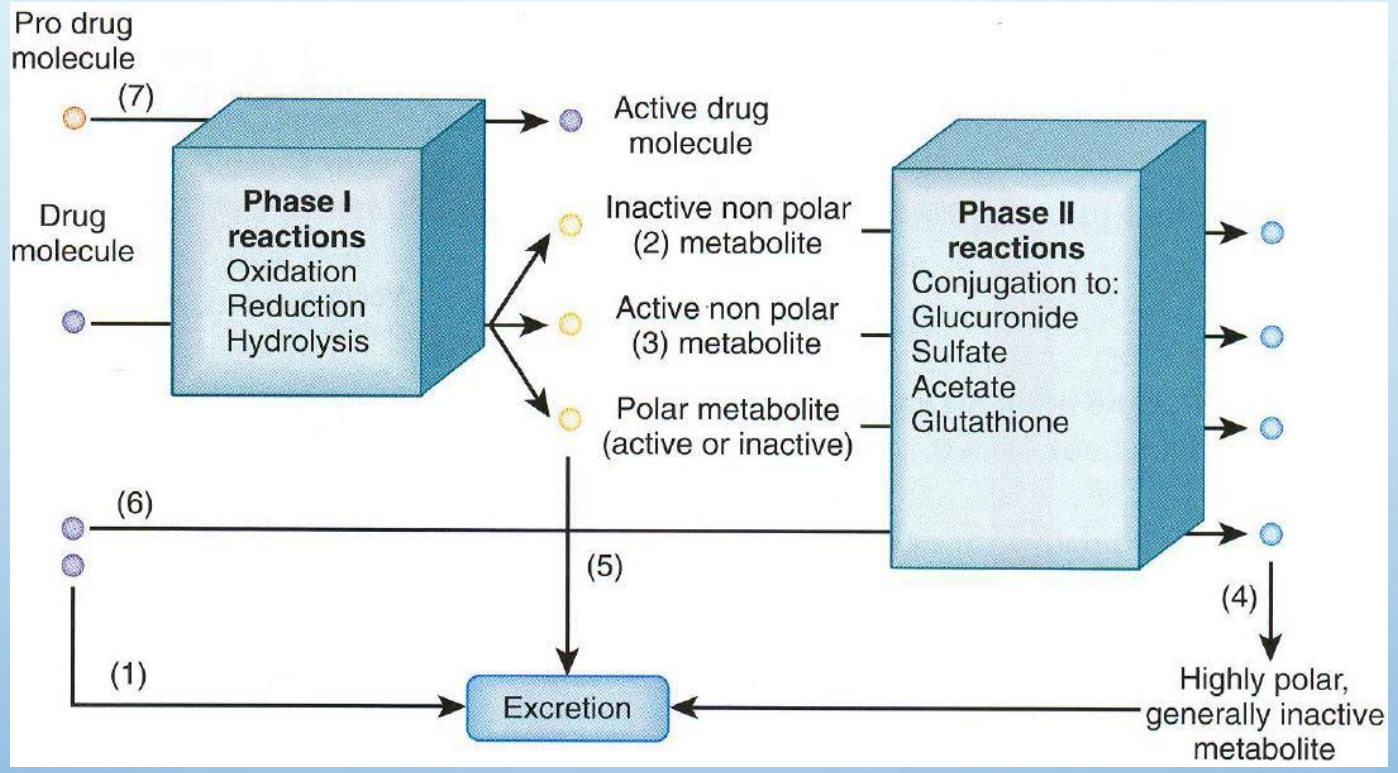
- Attaching polar or ionized functional groups to hydrophobic molecules:
  - hydroxyl (ionized)
  - amino (ionized)
  - phosphate (ionized)
  - sulfhydryl
  - aldehyde
  - ketone

2 PHASES: DRUG ELIMINATION



Rang/Dale/Ritter/Flower/Henderson, Pharmacology, Elsevier 2007

DRUG METABOLISM



Bardal/Waechter/Martin, Applied Pharmacology, Elsevier 2011

Cytochrome P450 (CYP) Catalytic Cycle

Heme-bound enzyme binds a substrate, activates oxygen, and converts the target molecule

1. Resting state (Fe^{3+})

CYP contains a heme group (protoporphyrin IX) with Fe^{3+} in the active site.

2. Substrate binds (Fe^{3+})

A substrate (drug, toxin, endogenous molecule) enters the active site and binds above the heme, displacing water.

3. First electron transfer (Fe^{2+})

An electron from NADPH (via cytochrome P450 reductase) reduces the heme iron from Fe^{3+} to Fe^{2+} (in the presence of substrate).

4. O_2 binds ($Fe^{2+}–O_2$)

Molecular oxygen (O_2) binds to the ferrous iron, forming a $Fe^{2+}–O_2$ complex (superoxo).

5. Second electron transfer and protonation ($Fe^{3+}–O_2^-$ / –OOH)

A second electron and protons convert the bound O_2 to a peroxo/hydroperoxo intermediate ($Fe^{3+}–O_2^-$ / Fe^{3+}–OOH).

6. Compound I forms (Fe(IV)=O + Por•+)

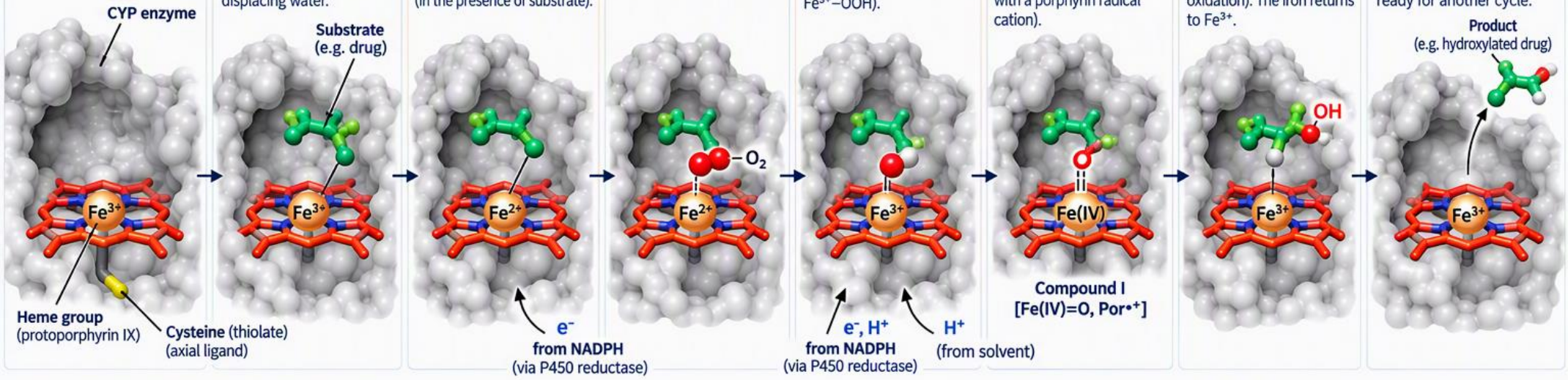
Protonation and O–O bond cleavage generate the highly reactive oxidizing species "Compound I" (Fe(IV)=O with a porphyrin radical cation).

7. Substrate oxidation (Fe^{3+})

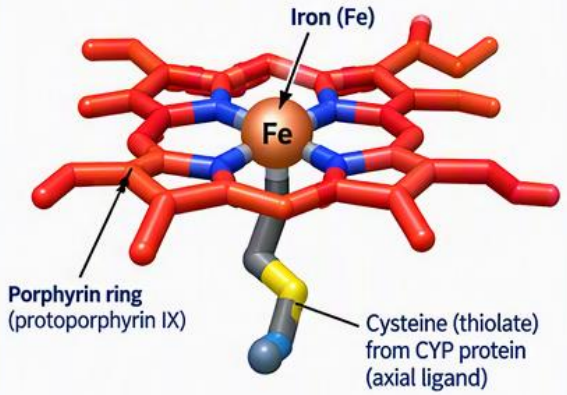
Compound I abstracts a hydrogen atom from the substrate, leading to hydroxylation (or other oxidation). The iron returns to Fe^{3+}.

8. Product release (Fe^{3+})

The oxidized product leaves the active site, water is released, and CYP returns to the resting Fe^{3+} state, ready for another cycle.



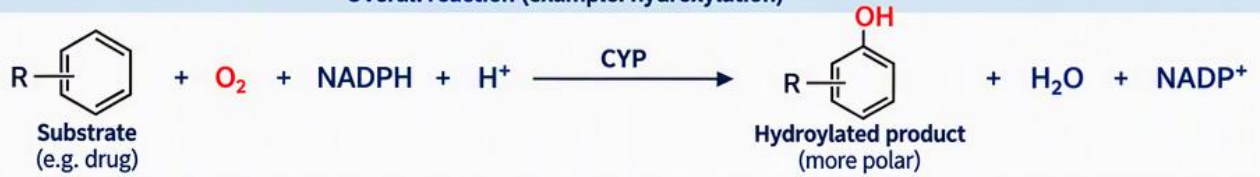
Close-up of the heme group in CYP



Summary of iron oxidation states during the CYP catalytic cycle

Fe^{3+} (resting) $\xrightarrow{+e^-}$ Fe^{2+} (ferrous) $\xrightarrow{+O_2}$ $Fe^{2+}–O_2$ (superoxo) $\xrightarrow{+e^-, H^+}$ $Fe^{3+}–O_2^-$ / –OOH (peroxo / hydroperoxo) $\xrightarrow{+H^+}$ Fe(IV)=O (Compound I) → Fe^{3+} (after product release)

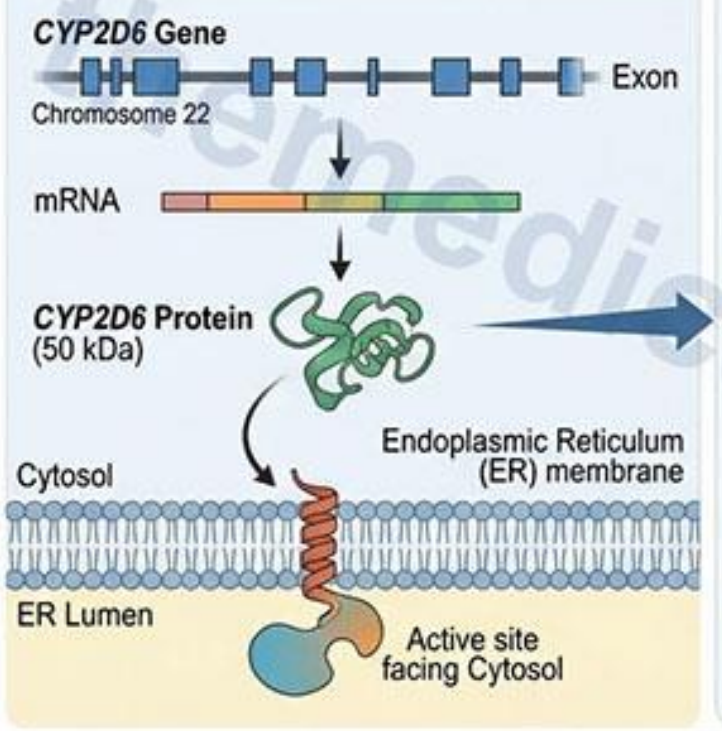
Overall reaction (example: hydroxylation)



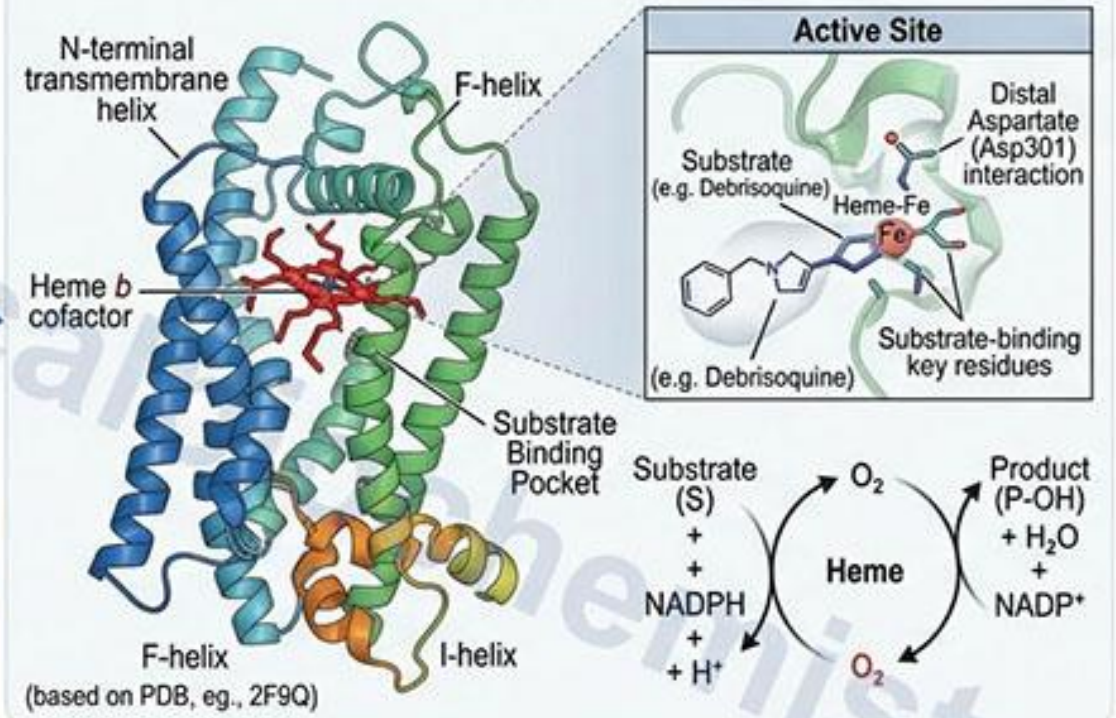
CYP enzymes use a heme iron (cycling between Fe^{3+} and Fe^{2+}) to activate oxygen and insert an oxygen atom into a wide range of substrates, helping to metabolize drugs, toxins, and endogenous compounds.

STRUCTURE, FUNCTION, AND PHARMACOGENOMICS OF CYP2D6 PROTEIN

A. GENE AND SUBCELLULAR LOCALIZATION



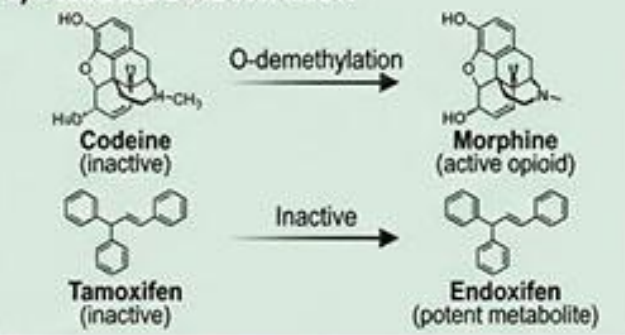
B. ENZYME STRUCTURE AND REACTION



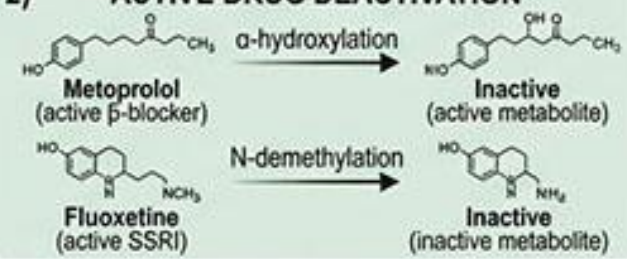
C. FUNCTIONAL ROLE: DRUG METABOLISM

CYP2D6 Metabolizes ~20-25% of Prescribed Drugs

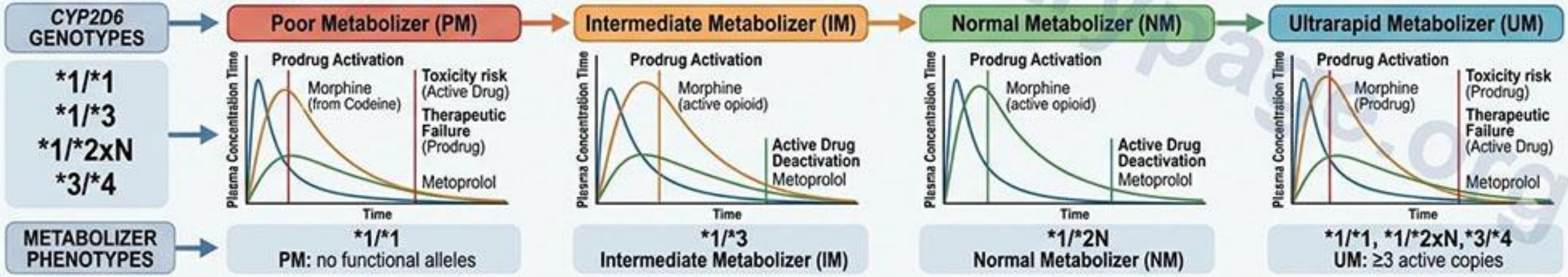
1) PRODRUG ACTIVATION



2) ACTIVE DRUG DEACTIVATION



D. IMPACT OF GENETIC POLYMORPHISM ON PHARMACOKINETICS



themedicalbiochemistrypage.org

https://themedicalbiochemistrypage.org/cytochrome-p450-cyp-enzymes/

DRUG METABOLISM

Phase I/II reactions mainly in liver

Phase I (functionalization), a.k.a. oxidation-reduction reactions

- Catabolic

  → products are usually more chemically reactive
  molecule becomes more polar + reactive:
  → facilitates excretion or further biotransformation

Phase II (conjugation)

- Anabolic

  → products are usually less chemically reactive

Decreased lipid solubility/increased water solubility
→ increased renal elimination

PHASE I – P450

P450 monooxygenase system

- cytochrome P450 enzymes: haem proteins
- 74 CYP gene families,
  3 main ones (CYP1, CYP2 and CYP3) involved in drug metabolism in human liver
- important variations in the expression and regulation of P450 enzymes (polymorphism)
- major interindividual variations in P450 enzymes in human populations: great importance in therapeutics

Isoenzyme P450	Drug(s)
CYP1A2	caffeine, paracetamol
CYP2B6	methadone
CYP2C9	ibuprofen
CYP2D6	codeine
CYP2E1	alcohol, paracetamol
CYP3A4,5,7	ciclosporine, nifedipine, indinavir, simvastatin

PHASE II - REACTION

- Involves conjugation of reactive groups (polar):
  - glucuronide
  - glutathione
  - sulfate
  - methyl
  - acetate
- Conjugation reaction can occur with:
  - directly the parent drug molecule
  - reactive intermediate generated by Phase I reactions
- resulting conjugate:
  almost always less pharmacologically active and less lipid-soluble
  → excretion to urine

- CAVE:
  Certain conjugates are more toxic!!!
  e.g. conversion of procainamide to N-acetylprocainamide
  (cyclophosphamide, halothane, methoxyflurane, paracetamol)

ENZYME INDUCTION

- drugs can **increase** ...
  - microsomal oxidase
  - conjugating systems

  ... activity when applied repeatedly:
  e.g. rifampicin, ethanol, carbamazepin

  → acceleration of substrate metabolism
  → decrease of pharmacologic action

- >200 drugs can cause enzyme induction
  → decrease of pharmacological activity of other drugs
  → development of tolerance

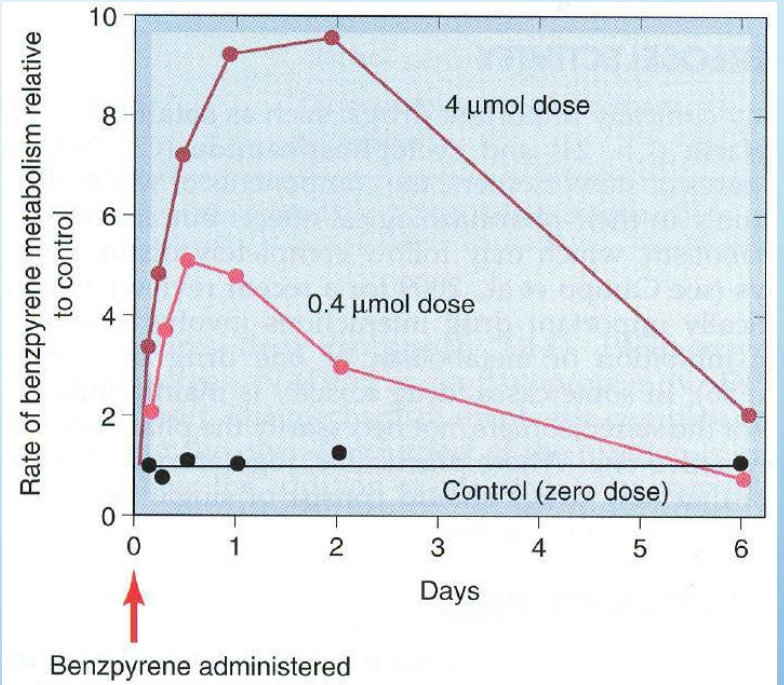


Fig. 9.4 Stimulation of hepatic metabolism of benzpyrene. Young rats were given benzpyrene (intraperitoneally) in the doses shown, and the benzpyrene-metabolising activity of liver homogenates was measured at times up to 6 days. (From Conney A H et al. 1957 J Biol Chem 228: 753.)

Rang/Dale/Ritter/Flower/Henderson, Pharmacology, Elsevier 2007

ENZYME INHIBITION

Inhibition of P450 enzymes:.
→ slows metabolism
→ prolonged effect duration of other drugs metabolized

Example

- Grapefruit juice reduces metabolism of:
  - terfenadine
  - ciclosporin
  - Ca^{2+}-channel antagonists

  → **Cave**: prescription of drug combinations

LIVER INDUCTION & INHIBITION OF CYP450

CYP	Substrates	Inducers	Inhibitors
1A2	Paracetamol, caffeine, phenacetin, tamoxifen, theophylline, warfarin	Smoking, charcoalbroiled foods, omeprazole	Galangin, fluvoxamine
2A6	Coumarin, tobacco nitrosamines, nicotine	Rifampin, Phenobarbital	Tranylcypromine, methoxsalen
2C9	Celecoxib, flurbiprofen, hexobarbital, ibuprofen, losartan, phenytoin	Barbiturates, rifampin	Tienilic acid, sulfaphenazole
2C19	Diazepam, naproxen, omeprazole, propanolol	Barbiturates, rifampin	Benzylphenobarbital, fluconazole
2D6	Bupranolol, clozapine, codeine, dextromethorphan, fluoxetine, haloperidol, metoprolol, oxycodone, paroxetine, risperidone, selegiline, tricyclic antidepressants.....	St.John´s wort, rifampin	Quinidine, paroxetine
3A4	Paracetamol, amiodarone, cisapride, cocaine, cortisol, cyclosporine, dihydroergotamin, diltiazem, erythromycin, ethinyl estradiol, gestodene, indinavir, lovastatin, methadone, ritonavir, tamoxifen, testosterone, cannabinol, verapamil....	Barbiturates, carbamazepine, glucocorticoids, macrolide antibiotics, pioglitazone, pheytoin	Diltiazem, erythromycin, fluconazole, grapefruit juice, itraconazole, ketoconazole, ritonavir

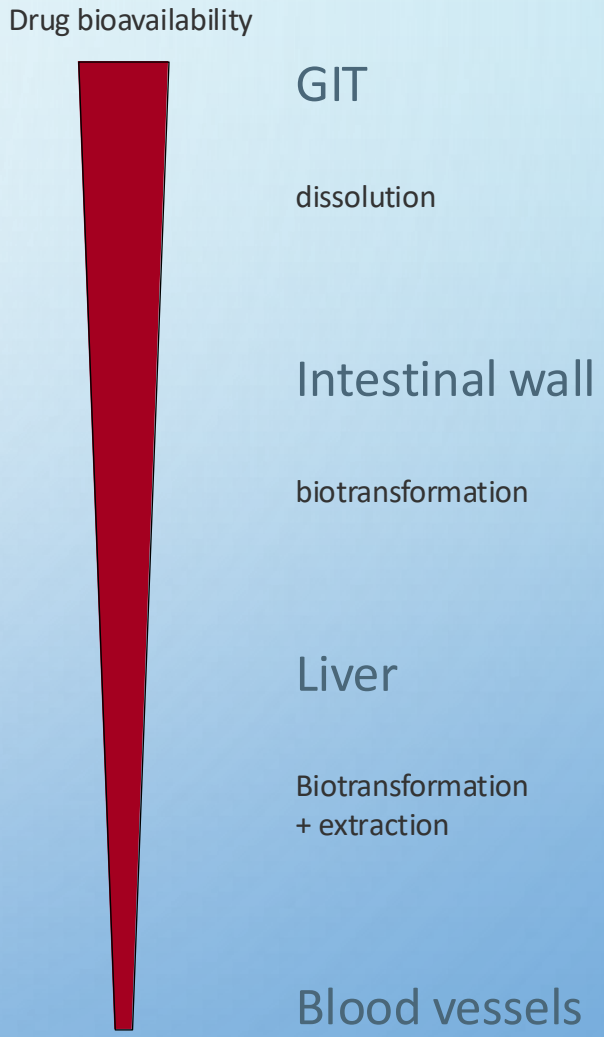
Cave: prescription drug interactions

FIRST-PASS EFFECT

- drug amount reaching systemic circulation
  ≠ drug amount absorbed
- **first pass effect**
  = drug amount metabolized during first passage through GIT wall
- First-pass elimination/reduced bioavailability after administration of
  e.g.
  - aspirin
  - lidocaine
  - morphine
  - verapamil
  - levodopa

→ *larger drug doses* is needed with oral administration than by other routes

→ *individual variations* in extent of first-pass metabolism can result in unpredictability with oral drugs intake



LEARNING OUTCOMES

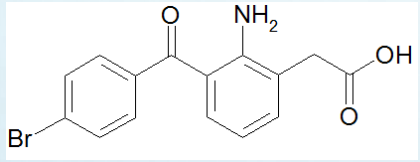
- Role of the liver in drug metabolism
- **Role of drugs in liver damage**
- Consequences of hepatic damage to drug metabolism & excretion

DRUG-INDUCED LIVER DAMAGE

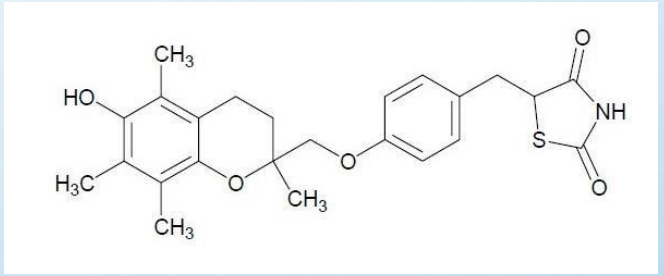
- 900 drugs, toxins and herbs:
  reported to cause liver injury
- drug-induced hepatic injury:
  most common reason for withdrawal of an approved drug
- in US, drugs account for >50% of acute liver failure
  (39% are due to paracetamol)
- early identification of drug related liver
  damage: decreases hepatotoxicity

→ **extensive knowledge of commonly implicated agents**

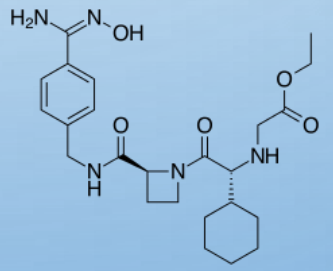
DRUGS WITHDRAWN FROM THE MARKET

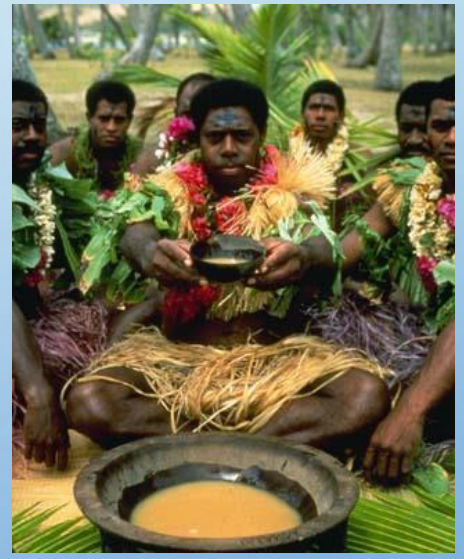


- **bromfenac** (NSAID)
  → induced severe cases of hepatic injury



- **troglitazone** (anti-diabetic agent)
  → induced severe cases of hepatic injury
- **kava kava** (tranquilizer)
  → liver failure in some people that used dietary supplements containing kava extract
- **ximelagatran** (anti-coagulant)
  → hepatotoxic side effects were reported





Medical Physiology & Experimental Pharmacology, 2018

Risk factors
drug-induced hepatotoxicity

- **Race**
  rate of metabolism controlled by P450 enzymes → can vary between individuals
- **Age**
  decreased clearance, drug to drug interactions, reduced hepatic blood flow in elderly persons → increased risk of hepatic injury
- **Sex**
  hepatic drug reactions more common in females, reasons unknown (due to body composition?)
- **Alcohol ingestion**
  alcohol induces liver injury and cirrhotic changes that alter drug metabolism
- **Liver disease**
  total CYP-450 is reduced, modification of doses should be based on knowledge of specific enzyme involved in metabolism

Pathophysiology & mechanisms of drug toxicity

Hepatocytes

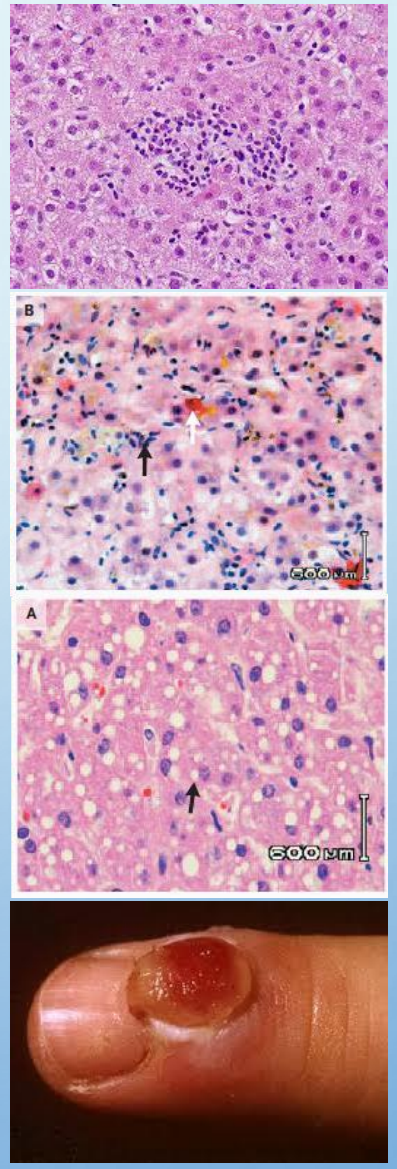
- exposed to high concentrations of metabolites
  → toxic concentrations can cause necrosis and programmed cell death
- drug-induced cell damage usually caused by reactive metabolites of the drug involving **non-covalent** and **covalent interactions** with target molecule

Hepatotoxicity includes hepatocellular and extracellular mechanisms

- **disruption of hepatocyte**
  covalent binding of drug to intracellular proteins → ↓ ATP level → actin disruption → degradation of actin fibrils causes rupture of membrane
- **Cytolitic T cell activation**
  covalent binding of drug to P-450 enzyme acts as immunogen → activation of T-cells, cytokines → multifaceted immune response
- **apoptosis of hepatocytes**
  activation of TNF-alpha receptor of Fas → programmed cell death

DRUG-INDUCED LIVER INJURIES

- **Necrosis**
  confined to particular zone of liver, can lead to severe disturbance in liver function → acute liver failure
- **Hepatitis (B)**
  hepatocellular necrosis associated with infiltration of inflammatory cells
  black arrow: acute hepatitis
  white arrow: hepatocellular necrosis
- **Steatosis (A)**
- hepatotoxicity manifests as triglyceride accumulation → can lead to fatty liver
- **Granuloma**
  usually associated with granulomas in other tissues, features of systemic vasculities and hypersensitivity



V.J. Navarro & J.R. Senior,
N Engl J Med 2006;354:731-9

DRUG TOXICITY MECHANISMS

Adverse drug reactions classified as:

- Type A (intrinsic or pharmacological)
- Type B (idiosyncratic)

Type A

- Predictable dose-response curves and well characterized toxicity mechanisms
- Injury occurs shortly after toxicity threshold is reached

Type B

- Non-predictable hepatotoxicity that is not related to dose
- Variable latency period
- Led to withdrawal of several drugs even after rigorous clinical testing

DRUG INVOLVED IN LIVER DAMAGE

- **isoniazid** (anti-tuberculotic drug)
  acetylation leads to reactive species responsible for the toxicity;
  severe hepatotoxicity in 1-2 % of patients
- **erythromycin** (antibiotic)
  may cause hepatic dysfunction, including hepatocellular and/or cholestatic
  hepatitis, with or without jaundice
- **allopurinol** (used in gout)
  few cases with allopurinol-induced liver injury
  → allopurinol-induced granulomatous hepatitis
- **fly agaric** (amanita muscaria)

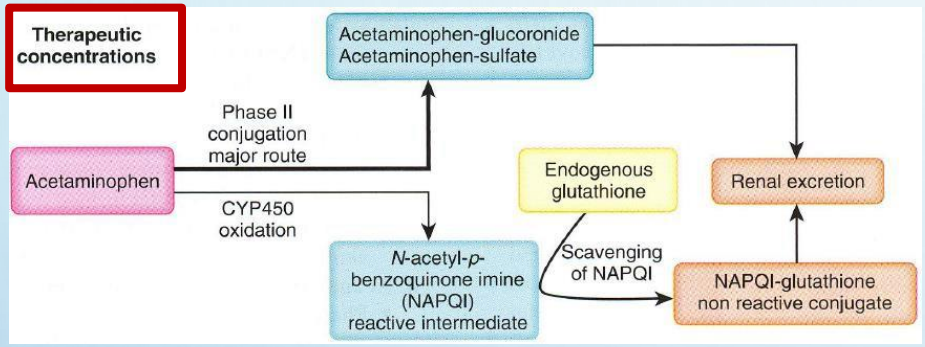


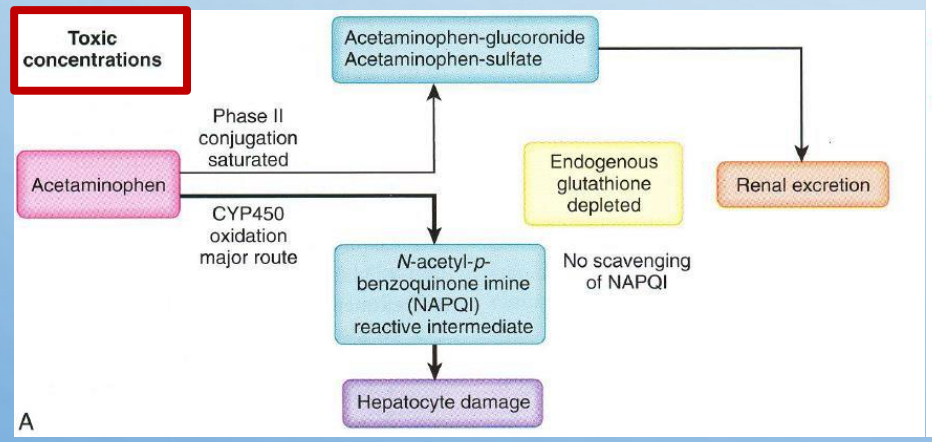
- **paracetamol (acetaminophen)**

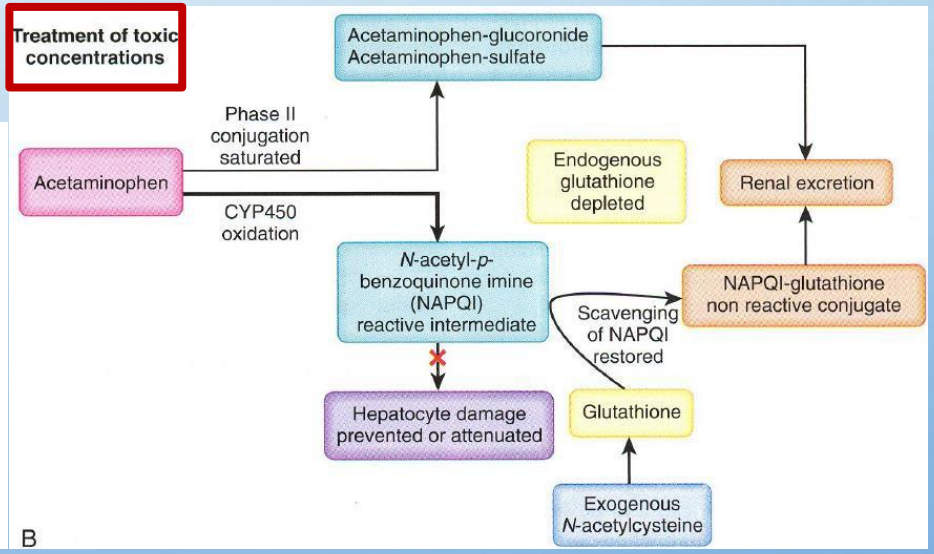
PARACETAMOL METABOLISM

- quite safe in therapeutic doses (1.2 g/d for adults)
- normally **glucuronidation** and **sulfation** to conjugates
- if paracetamol intake far exceeds therapeutic doses (10-15 g), *glucuronidation* and *sulfation* pathways are saturated
  → alternative P450-dependent pathway becomes important
- as long as glutathione (GSH) is available: little or no hepatotoxicity
  (agents that increase GSH: acetylcysteine i.v., methionine i.v.)
- Depletion of hepatic GSH:
  → accumulation of reactive toxic metabolites
  → react with nucleophilic groups of cellular proteins
  → hepatotoxicity

PARACETAMOL METABOLISM

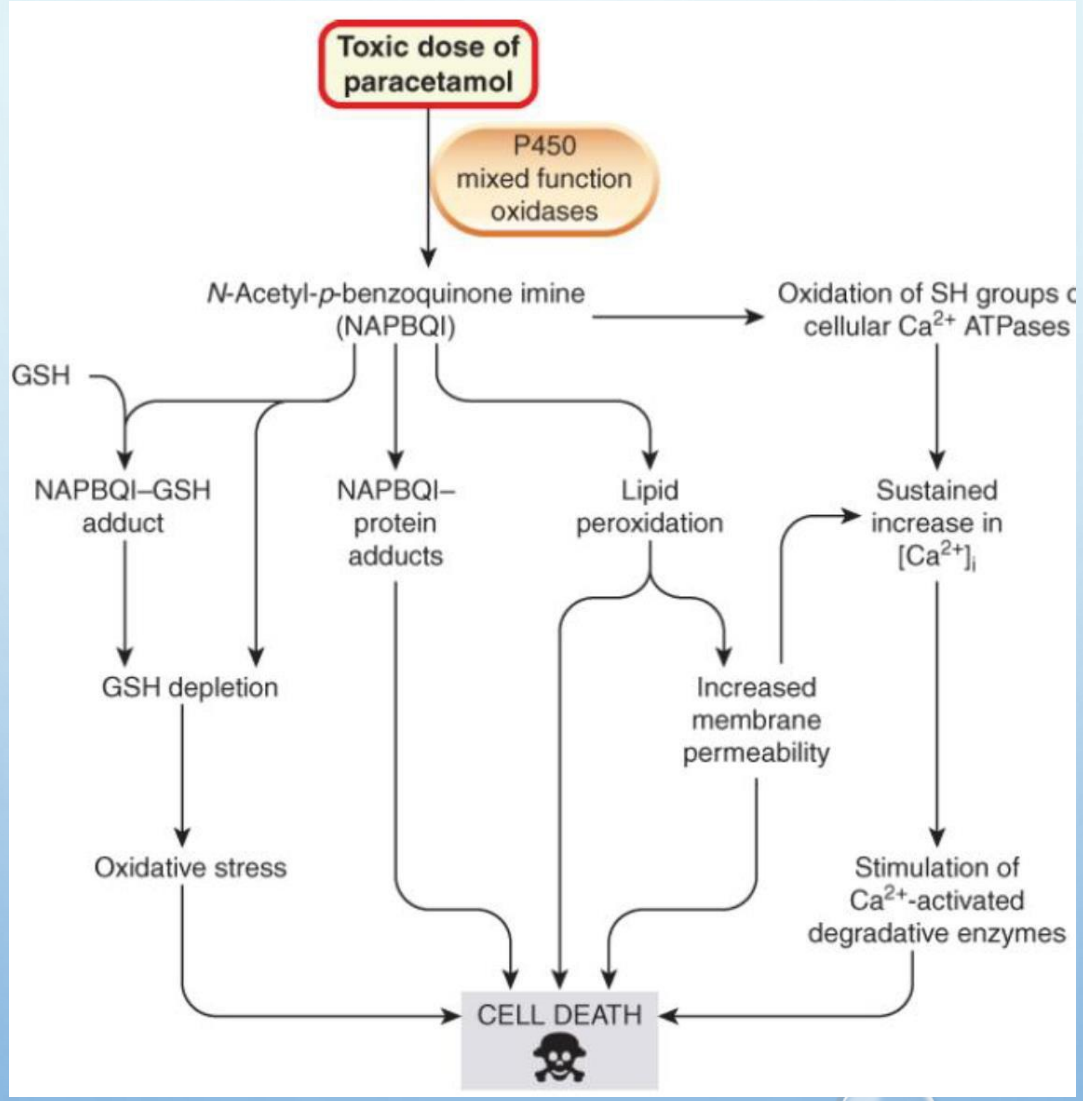






Bardal/Waechter/Martin, Applied Pharmacology, Elsevier 2011

PARACETAMOL METABOLISM



Rang/Dale/Ritter/Flower/Henderson, Pharmacology, Elsevier 2007

PARACETAMOL POISONING

- one of the drugs most commonly involved in **suicide** and **accidental poisonings** (7 g considered potentially toxic for adults)
- initial symptoms of acute paracetamol poisoning: nausea, vomiting
- hepatotoxicity occurs 24-48 hours later, in severe cases fulminant liver failure
  → hepatic encephalopathy and death
- antidote acetylcysteine (glutathione substitute):
  can prevent liver damage when given early within 8-10 hours

LEARNING OUTCOMES

- Role of the liver in drug metabolism
- Role of drugs in liver damage
- **Consequences of hepatic damage to drug metabolism & excretion**

CONSEQUENCES OF HEPATIC DAMAGE

- can be indicated by
  - ↑ aminotransferase (ALT, AST)
  - ↑ alkaline phosphatase (ALP)
  - ↑ total bilirubin levels

- increase of far more than 3 times upper limit does **not** necessarily lead to clinically significant liver damage
  → great capacity of liver to heal injury

+

- vague symptoms:
  fatigue, anorexia, nausea, discomfort in right upper quadrant, dark urine

↓

symptoms + biochemical evidence

=

drug-related hepatotoxicity should be considered

- **NORMAL LEVELS OF THE DIFFERENT ENZYMES:**

ALT (ALANINE TRANSAMINASE)	7 TO 55 UNITS PER LITER (U/L)
AST (ASPARTATE AMINOTRANSFERASE)	8 TO 48 U/L
ALP (ALKALINE PHOSPHATASE)	45 TO 115 U/L
ALBUMIN	3.5 TO 5.0 GRAMS PER DECILITER (G/DL)
TOTAL PROTEIN	6.3 TO 7.9 G/DL
BILIRUBIN	0.1 TO 1.2 MILLIGRAMS PER DECILITER (MG/DL)

DRUG-INDUCED LIVER INJURIES

- **Necrosis**
  confined to particular zone of liver, can lead to severe disturbance in liver function → acute liver failure
- **Hepatitis**
  hepatocellular necrosis associated with infiltration of inflammatory cells
- **Steatosis**
  hepatotoxicity manifests as triglyceride accumulation → can lead to fatty liver
- **Granuloma**
  usually associated with granulomas in other tissues, features of systemic vasculities and hypersensitivity

Medical Physiology & Experimental Pharmacology, 2018

HEPATIC DAMAGE

Consequences to drug metabolism and excretion?

~~**Metabolism converts lipophilic drugs into more water-soluble compounds → essential for excretion in urine**~~

- **prolonged duration of drug action**
- **detoxification process inhibited**
- **higher concentration of toxic substrates**
- **reduced drug excretion via kidneys**

CONSEQUENCES OF HEPATIC DAMAGE

~~Inactivation of various substances, e.g. toxins, steroids, other hormones~~

- prolonged activity of toxic and non-toxic metabolites can lead to further damage
- inhibition of detoxification processes
  → careful prescription of drugs that are metabolized via liver enzymes

CONSEQUENCES OF HEPATIC DAMAGE

~~Drug-metabolizing enzymes convert pharmacologically inactive prodrugs to active molecules~~

- reduced or no effect of certain drugs and substrates, e.g. cyclophosphamide

CONSEQUENCES OF HEPATIC DAMAGE

~~Liver involved in synthesis of endogenous substrates, cholesterol, steroid hormones, vitamin D, bile acids.~~

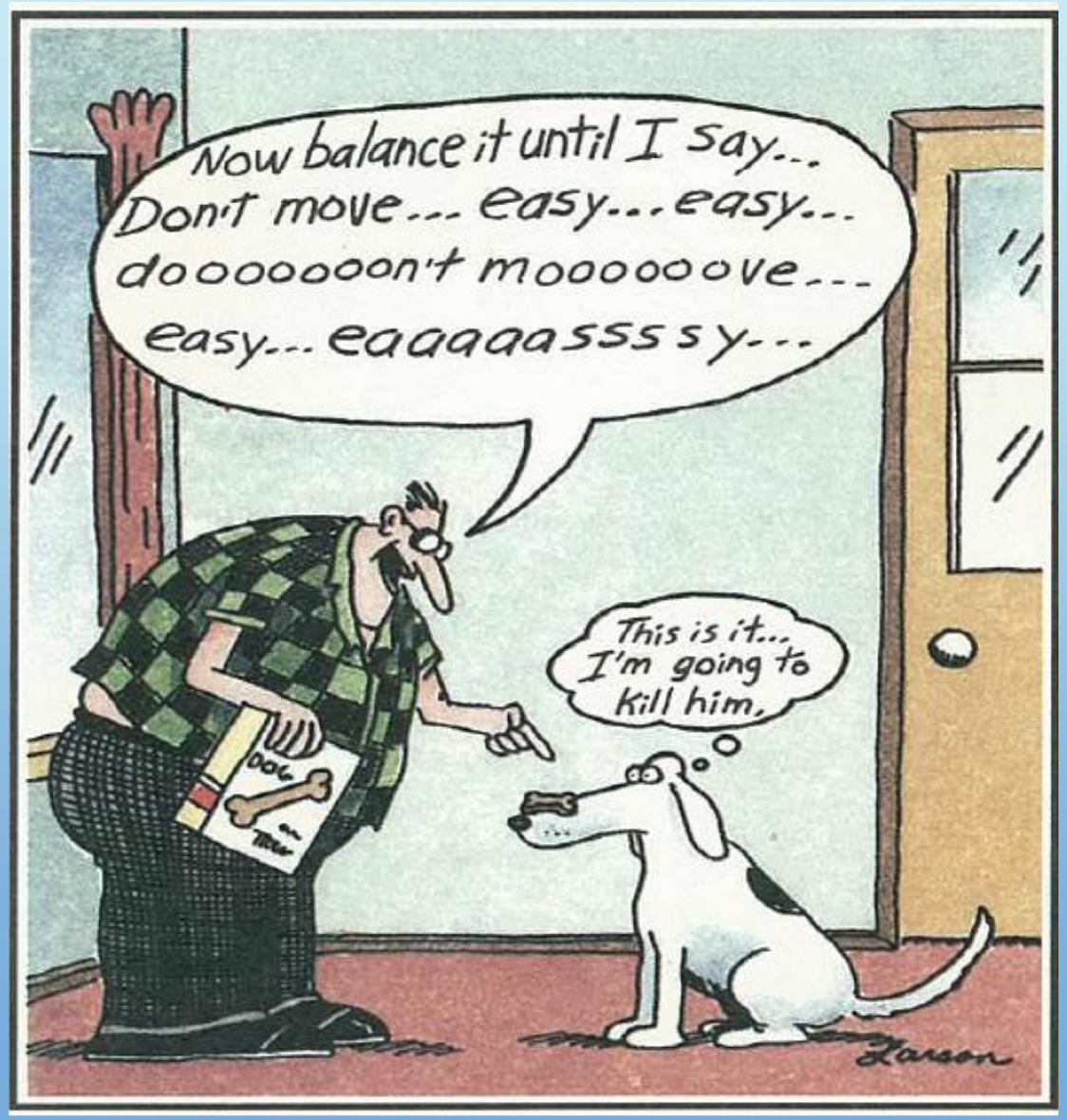
- reduction of proper membrane permeability and fluidity
- interference with cardiovascular, metabolic, immunologic, homeostatic and reproductive functions
- interference in calcium and phosphorus blood levels → impairment of bone growth and remodeling
- impaired dietary fat processing

Literature

- BASIC AND CLINICAL PHARMACOLOGY, B.G. KATZUNG, 10TH ED.
- ANATOMY & PHYSIOLOGY, JENKINS/KEMNITZ/TORTORA,WILEY 2010
- PHARMACOLOGY, RANG/DALE/RITTER/FLOWER/HENDERSON, ELSEVIER 2007
- ARZNEIMITTELWIRKUNG, MUTSCHLER, 8TH ED.
- DRUG-INDUCED HEPATOTOXICITY,W.M. LEE, N ENGL J MED 2003;349:474- 85
- DRUG-RELATED HEPATOTOXICITY,V.J. NAVARRO & J.R. SENIOR, N ENGL J MED 2006;354:731-9

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Questions? Comments?



PHASE I - ENYZMES

Oxidation

- Large proportion of drug metabolism (reduction & hydrolysis less common)
- Primarily monooxygenases, microsomal mixed function oxygenases within ER:
  - Cytochrome P450 (CYP)
  - flavin monooxygenase family
  - hydrolytic enzymes (e.g. epoxide hydrolase)
- P450: >80% drug oxidation

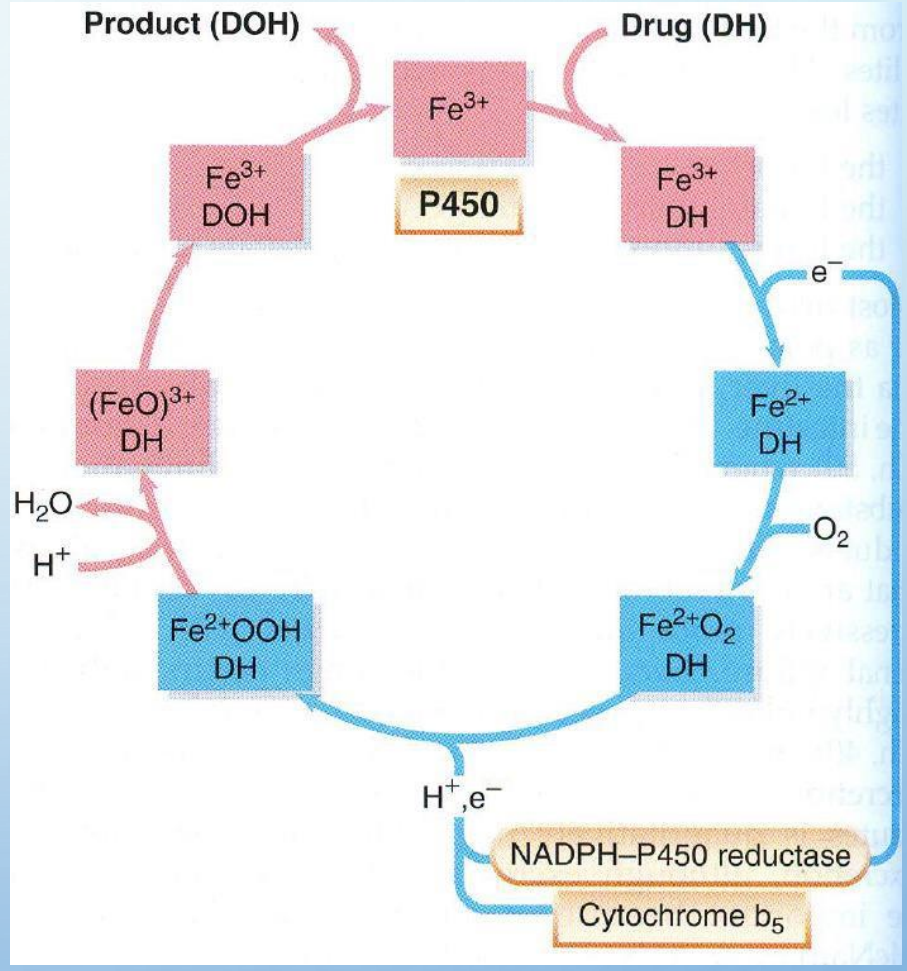
Reduction

- nitroglycerin (→angina)

Hydrolysis

- hydrolase (epoxide hydrolase)
- esterase (cholinesterase)

Medical Physiology & Experimental Pharmacology, 2018



pink: iron in ferric state within CYP
blue: iron in ferrous state within CYP

Rang/Dale/Ritter/Flower/Henderson, Pharmacology, Elsevier 2007